

Seismic noise study for accurate *P*-wave arrival detection via MODWTAli G. Hafez^{a,b,*}, Mostafa Rabie^c, T. Kohda^d^a LTLab Inc. Kyushu University, Higashi-ku, Hakozaki 6-10-1, Fukuoka-shi 812-8581, Japan^b National Research Institute of Astronomy and Geophysics, (NRIAG), Helwan 11722, Egypt^c Department of Electrical Engineering, Faculty of Engineering, Minia University, Minia, Egypt^d Department of Informatics, Kyushu University, 744 Motoooka, Nishi-ku, Fukuoka-shi 812-8581, Japan

ARTICLE INFO

Article history:

Received 5 August 2012

Received in revised form

2 December 2012

Accepted 3 December 2012

Available online 10 December 2012

Keywords:

Automatic picking

P-wave arrival

MODWT

Wavelets

Earthquakes

ABSTRACT

The arrival timing of the onset of microseisms and weak events is difficult to be picked even manually. The proposed algorithm uses the maximal-overlap discrete wavelet transform (MODWT) to perform manual detection for such weak events. A seismic noise analysis was done to choose the best criteria for showing clear *P*-wave arrival. This algorithm is also used as an accurate automatic *P*-wave picking algorithm. The noise level at a seismic station does not affect the proposed picking algorithm because it adapts itself to the noise level in front of each earthquake. Local events recorded by the Egyptian National Seismic Network (ENSN) were used to test this proposed algorithm. The overall average error was found to be 0.02 s.

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1. Introduction

P-wave picking is the first step in seismic signal analysis. Accurate determination of onset arrival timing is a very important issue in the correct calculation of earthquake magnitude, location, timing and other parameters that can be used for building catalogues and making more accurate seismological maps. Automatic trigger algorithms are relatively ineffective when compared to a seismologist's pattern recognition ability to read seismograms, which is based on years of experience. However manual analysis of seismograms and phase picking is time-consuming and tedious. The rapid increase in the size of seismogram databases has created an ever increasing need for reliable automatic picking methods in seismological projects involving medium- to large-size datasets. Automatic onset picking is of great importance in acquisition of triggered seismic data, event identification, event location, early warning systems, seismic travel-time tomography and source mechanism analysis.

In addition to the large amount of digital data available at observatories, the *P*-wave has many shapes, which makes it more difficult to design a picking algorithm for each of these shapes. Of these shapes, some are sharp and abrupt as they have high power and frequency content, others are with gradual increases in power and/or frequency and another type which is the weak *P*-wave which has low power and frequency content, which is the most difficult to

determine, even sometimes manually. There is another classification for the types of *P*-wave which is made according to the layer which the *P*-wave was refracted from. For example, *P_g* is the *P*-wave which is refracted from granitic layer and *P_n* is the *P*-wave which is refracted from mantle. Motivated by these facts, many studies have been conducted to improve the performance of automatic algorithms as was this study.

Various algorithms have been proposed for automatic detection of onset arrival. Earle and Shearer (1994) and Tong and Kennett (1996) used simple amplitude and energy threshold, Wagner and Owens (1996) used the principal eigenvalue in the frequency domain as a detector.

Withers et al. (1998) made a review paper in which they had systematically compared several trigger algorithms. Sleeman and Van Eck (1999) used a large dataset of 1109 events which is the same dataset used by Baer and Kradolfer (1987). Sleeman and van Eck were able to increase the number of correct picks to 71%, but they had to use different band-pass filters to make signal enhancement. Such enhancement is not needed in our algorithm, even though we used the same range of event magnitudes used by Sleeman and Van Eck. Sophisticated pattern recognition and adaptive and artificial neural network-based approaches have been proposed by Aki and Hiroaki (2003). Colak et al. (2005) and Anant and Dowla (1997) used wavelet methods on three component stations, while Zhang et al. (2003) used multi-scale wavelet analysis for single-component recordings.

Sophisticated adjustments of operational parameters of these algorithms to actual signals and seismic noise at each site

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motivated Hafez et al. (2009, 2010) to introduce two parameter free algorithms based on spectrogram and multi resolution analysis (MRA) of Haar wavelets, respectively. A new parameter free algorithm based on MRA of maximal overlap discrete wavelet filters is introduced in this paper. This proposed algorithm is used to support manual detection of the onset of weak events and automatic detection of *P*-wave of events that have local magnitudes around 1, where such low local magnitude earthquakes have weak *P*-waves which sometimes cannot be analyzed manually.

To perform accurate *P*-wave picking, we studied the characteristics of the seismic noise in which the *P*-wave exists. From the results of this study, features were extracted that discriminate between seismic noise and events. These features were used to develop the algorithm introduced in this research. Seismic noise is defined as the recorded waveform when no event is recorded. This noise, however, is not useless and carries a wealth of information. Many studies have been done to extract some of this information. Withers et al. (1996) used a deep (1500 m) cased borehole near the town of Datil in west central New Mexico to study the relation between high-frequency (> 1 Hz) seismic noise characteristics and wind speed. Cara et al. (2003) analyzed seismic noise recorded in the Colfiorito area to determine the

micro tremor reliability in fitting the resonance frequency observed during earthquakes. Weaver (2005) reported many applications extracted information from seismic noise. Burtin et al. (2008) did spectral analysis of seismic noise to find the influence of rivers on the amount of seismic noise and its spatiotemporal variation.

Maximal overlap discrete wavelet transform (MODWT) is a modified version of the discrete wavelet transform (DWT), Percival and Walden (2000). MODWT has been discussed in wavelet literature in the context of infinite sequences under the name “un-decimated DWT”, Shensa (1992), and in the context of “power of two” sequences under the names “stationary DWT” Nason and Silverman (1995), “translation invariant DWT” Coifman and Donoho (1995); Liang and Parks (1996) and “time invariant DWT” Pesquet et al. (1996). While the wavelet coefficients of the DWT can be interpreted as a difference between two weighted averages, the intervals over which these averages are made are rigidly fixed a priori and hence might not line up well with interesting features in a time series. A change in the starting point can result in quite different results due to the overlap of the time series with the averaging interval predefined by the DWT. As we shall see, the MODWT is an attempt to avoid the effects of the

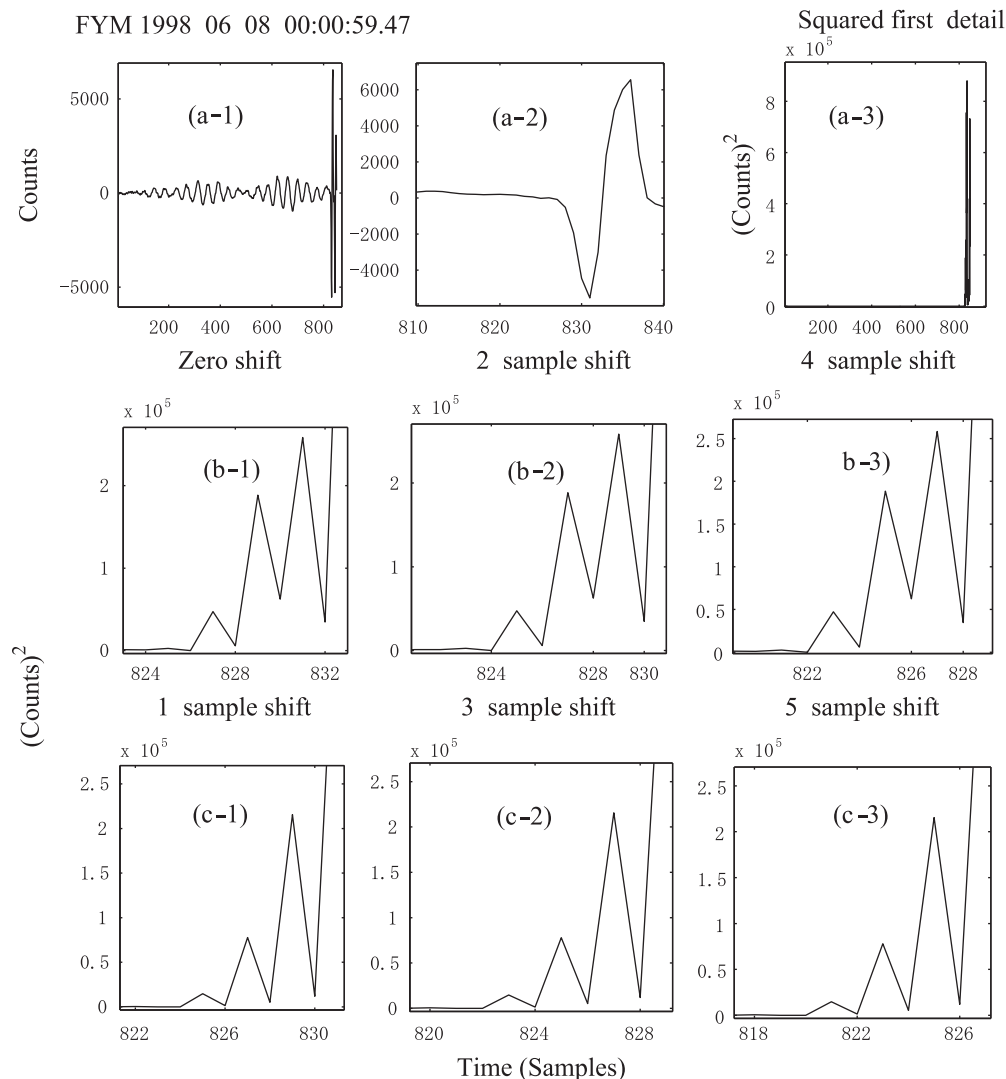


Fig. 1. Effect of the starting point choice on MRA of DWT using db8 wavelet filter. a-1 The seismic record in time domain as recorded at FYM station observed on June 8, 1998. a-2 Segment around *P*-wave of a-1 is enlarged. a-3 DWT squared first detail of a-1 using db8 wavelet filter. b-1, b-2 and b-3 Segments around *P*-wave of DWT squared first detail using db8 wavelet filter for a-1 with zero, two and four sample shifts, respectively. c-1, c-2 and c-3 Segments around *P*-wave of DWT squared first detail using db8 wavelet filter for a-1 with one, three and five sample shifts, respectively.

choice of a starting point by essentially including all possible placements of the averaging interval. Percival and Walden (2000) investigated the sensitivity of the DWT to the effect of the starting point in a comparison with the use of MODWT on a “Bump” time series. In this investigation, we show the effect on the DWT coefficients of shifting the original time series and the corresponding MRA details. Five important properties that distinguish the MODWT from the DWT are listed in Percival and Walden (2000).

2. Maximal overlap discrete wavelet transform (MODWT)

The motivation for formulating this MODWT was essentially to define a transform that acts as much as possible, like the DWT, but that is defined to any number of samples and does not suffer from the DWT’s dependence on the choice of the starting point of a time series as will be seen in the end of this section. This dependence is result of down sampling the outputs from scaling and wavelet filters at each stage, Percival and Walden (2000). In

order to remove the dependence to the starting point, it is necessary to eliminate this down sampling.

Filter outputs, which are usually discarded at the first stage when calculating DWT coefficients, can be obtained by applying the DWT calculations to the circularly shifted vector TX rather than X . Where T is an $N \times N$ circular shift matrix such that, if $X = [X_0; X_1; \dots; X_{N-2}; X_{N-1}]$, is an $N \times 1$ column vector, then, $TX = [X_{N-1}; X_0; X_1; \dots; X_{N-2}]$ and $T^{-1}X = [X_1; X_2; \dots; X_{N-2}; X_{N-1}; X_0]$.

The idea is to calculate the DWT coefficients twice, once to X and once to TX , then to merge the two sets of DWT coefficients to form the MODWT coefficients, the whole derivation behind this operation can be found in Percival and Walden (2000). The MODWT scaling and wavelet filters ($\tilde{g}_l$ and $\tilde{h}_l$, respectively) are related to the DWT filters with the relations $\tilde{g}_l = g_l/\sqrt{2}$ and $\tilde{h}_l = h_l/\sqrt{2}$. Note also that the quadrature mirror relationships between the DWT wavelet, h_l , and scaling, g_l , filters are also hold for the MODWT filters:

$$\tilde{g}_l = (-1)^{l+1} \tilde{h}_{L-1-l} \text{ and } \tilde{h}_l = (-1)^l \tilde{g}_{L-1-l}$$

where $l=0,1,2,\dots,L-1$ and L is filter width.

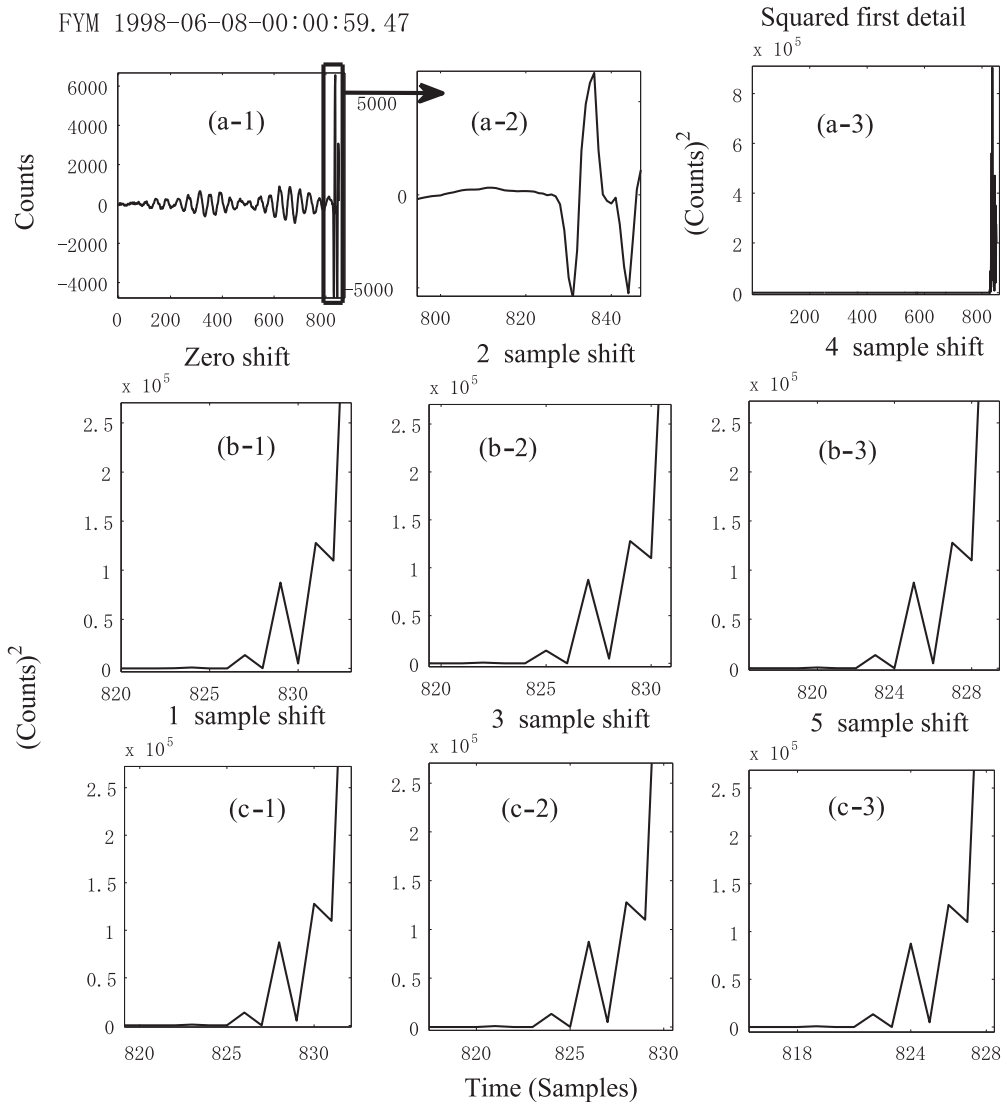


Fig. 2. Starting point choice does not affect MRA of MODWT, example; using db8 wavelet filter. a-1 The seismic record in time domain as recorded at FYM station observed on June 8, 1998. a-2 Segment around P-wave of a-1 is enlarged. a-3 MODWT squared first detail of a-1 using db8 wavelet filter. b-1, b-2 and b-3 Segments around P-wave of MODWT squared first detail using db8 wavelet filter for a-1 with zero, two and four sample shifts, respectively. c-1, c-2 and c-3 Segments around P-wave of MODWT squared first detail using db8 wavelet filter for a-1 with one, three and five sample shifts, respectively.

The description of all calculations necessary for finding the MODWT wavelet and scaling coefficients and its MRA can be found in Percival and Walden (2000). The n th element of the first stage scaling and wavelet coefficients are as follows,

$$\tilde{V}_{1,n} = \sum_{l=0}^{L-1} \tilde{g}_l x_{n-l \bmod N} \quad (1)$$

$$\tilde{W}_{1,n} = \sum_{l=0}^{L-1} \tilde{h}_l x_{n-l \bmod N} \quad (2)$$

where $x(n)$ is the input time series, $n = 0, 1, \dots, N-1$ and ' $a \bmod b$ ' stands for ' a modulo b '.

Details ($\tilde{A}_{j,n}$) and approximations ($\tilde{D}_{j,n}$) at each stage can be calculated from the MODWT coefficients of this stage. Details represent the high frequency components of this stage where approximations are the low frequency components.

Details are the original signal filtered within a specific band width that can be calculated from the equation $\frac{F}{2^{j+1}} \leq |f| \leq \frac{F}{2^j}$, where F is 50 Hz because the sampling rate is 100 samples per second and j is the stage number.

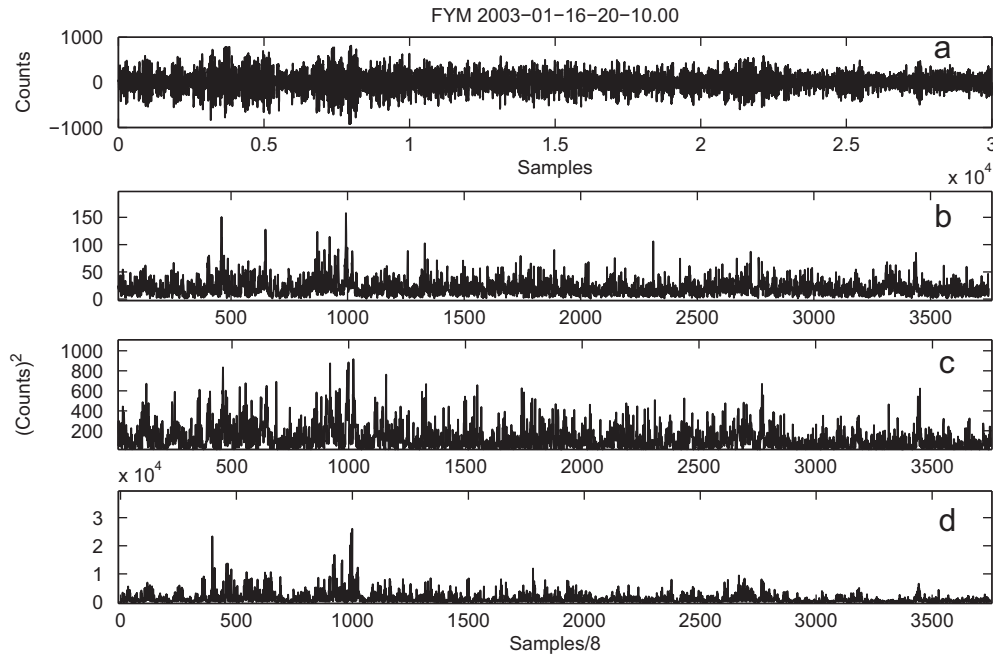


Fig. 3. The lowest noise level can be seen in the first detail (b). (a) Original time series of a noise segment recorded with the vertical component from the FYM station. (b) First MODWT squared detail of this noise segment using Haar wavelet filter. (c) Second MODWT squared detail. (d) Third MODWT squared detail.

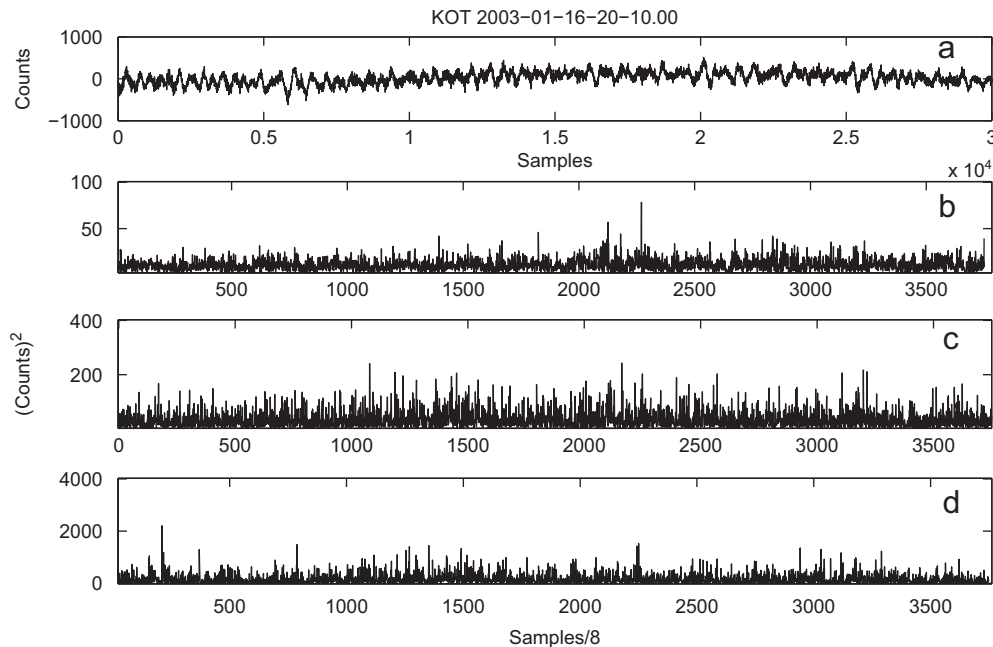


Fig. 4. The lowest noise level can be seen in the first detail (b). (a) Original time series of a noise segment recorded with the vertical component of the KOT station. (b) First MODWT squared detail of this noise segment using Haar wavelet filter. (c) Second MODWT squared detail. (d) Third MODWT squared detail.

The first stage approximation and detail can be calculated by:

$$\tilde{A}_{1,n} = \sum_{l=0}^{L-1} \tilde{g}_l \tilde{V}_{1,n+l \bmod N} \quad (3)$$

$$\tilde{D}_{1,n} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{W}_{1,n+l \bmod N} \quad (4)$$

The frequency content of the checked three details (first, second and third) are 12.5–25, 6.25–12.5 and 3.125–6.25 Hz, respectively.

By applying the same sequence for the DWT described in Hafez et al. (2010), we will have the n th element of the j th stage MODWT scaling, wavelet coefficients $\tilde{V}_j$, $\tilde{W}_j$ and approximation, $\tilde{A}_j$, and detail, $\tilde{D}_j$, which is illustrated by the following equations:

$$\tilde{V}_{j,n} = \sum_{l=0}^{L_j-1} \tilde{g}_{j,l} \tilde{x}_{n-l \bmod N} \quad (5)$$

$$\tilde{W}_{j,n} = \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} \tilde{x}_{n-l \bmod N} \quad (6)$$

$$\tilde{A}_{j,n} = \sum_{l=0}^{N-1} \tilde{g}_{j,l} \tilde{V}_{j,n+l \bmod N} \quad (7)$$

$$\tilde{D}_{j,n} = \sum_{l=0}^{N-1} \tilde{h}_{j,l} \tilde{W}_{j,n+l \bmod N} \quad (8)$$

where $\{\tilde{g}_l\}$ is $\{\tilde{g}\}$ periodized to length N and $\{\tilde{h}_l\}$ is $\{\tilde{h}\}$ periodized to length N . A detailed definition of the periodized filters can be found in Percival and Walden (2000).

The original time series can be decomposed from the MRA of the MODWT as follows:

$$x(n) = \tilde{D}_1 + \tilde{D}_2 + \dots + \tilde{D}_j + \tilde{A}_j. \quad (9)$$

2.1. Effect of choice of start point on DWT coefficients

In this section, effect of the choice of the starting point on DWT coefficients is investigated. To perform this investigation, Daubechies wavelet of order 8 (abbreviated as db8) is used to calculate MRA first detail of both DWT and MODWT.

Let $x_n = x_0 + x_1 + x_2 + x_3 + \dots + x_N$ and $x_{n,t} = x_t + x_{t+1} + x_{t+2} + \dots + x_N$ where x_n is the original time series of length N and $x_{n,t}$ is x_n shifted by t samples. In this investigation t will take values from zero to five. To perform this investigation of the dependence on the choice of the starting point, two steps are done as follows.

First step: squared first details of both DWT and MODWT of x_n are calculated. DWT first detail can be calculated using Eq. (8) in Hafez et al. (2010).

Second step: squared first details of both DWT and MODWT of $x_{n,t=0:5}$ are also calculated. These two steps are made on earthquake recorded at FYM station on June 8, 1998 using db8 wavelet filter. Fig. 1, illustrates the seismic record at FYM. Squared first detail of MRA of DWT of this record is calculated at zero shifting using db8 wavelet filter (a-3). (a-2) is the enlarged window around the P -wave arrival. Performing zero to five sample shift to the original time series, (plotted in a-1), then calculating squared first detail of each shifted version and zooming on the P -wave arrival, will generate six details as shown in Fig. 1 which are (b-1, b-2, b-3, c-1, c-2 and c-3). The six details are arranged

Table 1

MODWT squared detail standard deviation values of the FYM, HAG and KOT stations.

Detail	MODWT squared detail standard deviation (counts) ²		
	FYM	HAG	KOT
First	36.67	7.95	77.17
Second	146.96	20.998	329.77
Third	808.81	59.84	248.52

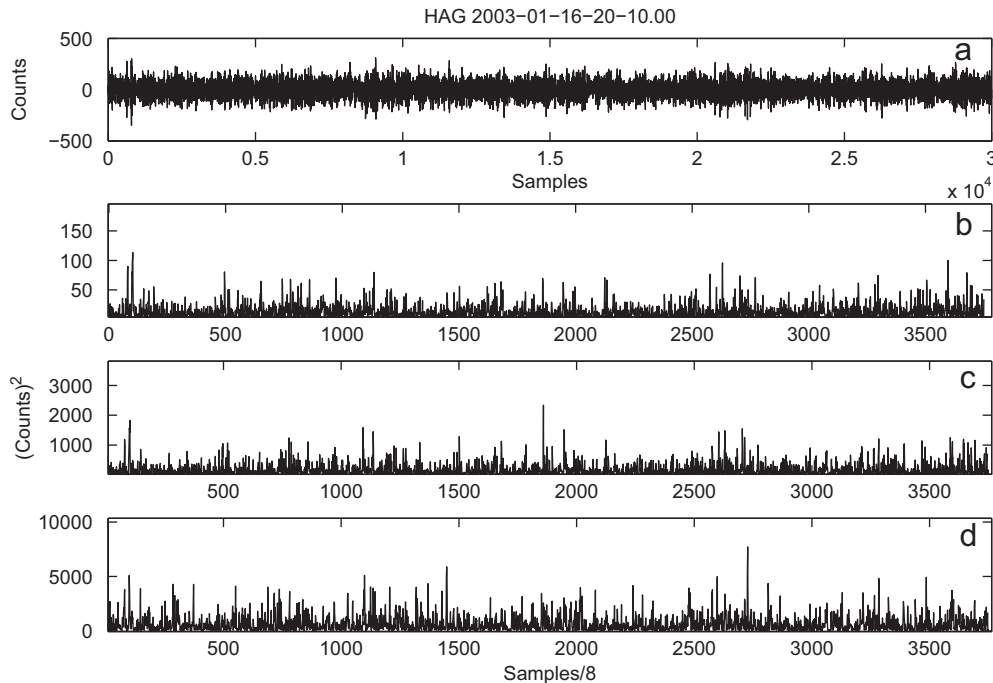


Fig. 5. The lowest noise level can be seen in the first detail (b). (a) Original time series of a noise segment recorded with the vertical component of the HAG station. (b) First MODWT squared detail of this noise segment using Haar wavelet filter. (c) Second MODWT squared detail. (d) Third MODWT squared detail.

with in the figure in two rows as follows: zero, two and four sample shift details are in the second row; one, three and five sample shift details are in the third row. Inspecting these focused *P*-wave views, we can recognize that there are two different patterns, one in the second row subfigures and the other in the third row. The differences are in both amplitude and shape of *P*-wave arrivals between second and third row details which proves that the choice of the starting point affects DWT details.

On the other hand the same event, recorded at FYM station, is checked with db8 wavelet filter to determine the squared first details of MRA of MODWT as shown in Fig. 2. From this figure we can recognize that the shift of the starting point does not affect either amplitude or shape of *P*-wave arrival determined by MRA of MODWT.

3. Seismic noise study

Seismic noise, a large part of the seismogram, can be generated by nature, weather or industry. Researchers who deal with seismic data, often first need to analyze the ambient seismic noise recorded at the stations under inspection. As the target of the proposed algorithm is to detect the first phase of earthquakes,

which are frequently buried in seismic noise, similarly, we did a study on seismic noise using MODWT. From this study we can select the criteria for optimum *P*-wave arrival picking. The changing behavior of seismic noise demands a year round campaign that covers all different activities. Noise measurements are chosen according to the following considerations:

1. We used seismic noise recorded by the vertical component of FYM, HAG and KOT stations belonging to the Egyptian National Seismic Network (ENSN) because the events to be checked were recorded at these stations.
2. Type of seismometer mounted at site; where the FYM and HAG stations have three component short period seismometers and the KOT station has a very broadband seismometer. Appendix A1 gives some notes about the classification of the seismometer.
3. Noise measurements for the three stations were taken four times per day. From each week of the four weeks of a month, only one day is chosen. That means four days per month which will count for sixteen records per month. Records were taken to cover a whole year. Samples were taken in one month for every season of the year to account for seasonal variation (64 segments per station, 192 segments total).

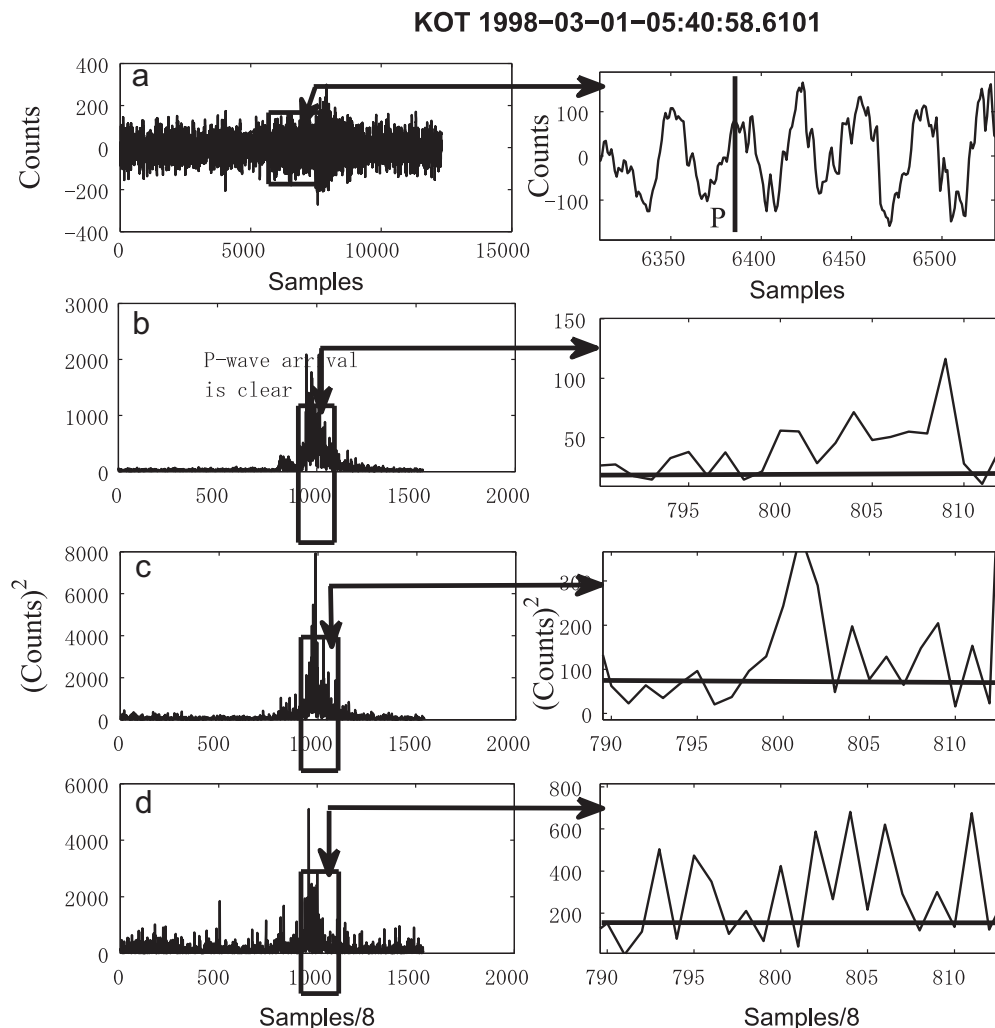


Fig. 6. A typical weak event recorded at KOT station. We can see that the amplitudes of the first detail are greater than the threshold over a certain period. (a) Complete weak event and the enlarged window at onset of the original time series where the *P*-wave arrival time is marked as P. (b) Squared complete first stage detail. (c) Squared complete second stage detail. (d) Squared complete third stage detail. The bold horizontal line in (b, c and d) is the average noise level recorded at the start of each segment.

4. The timing of the records was also chosen to avoid events by reviewing the bulletin published by [National Research Institute of Astronomy and Geophysics \(NRIAG\) \(2003\)](#).
5. Each noise segment is five minutes long with a sampling rate of 100 samples per second.

Figs. (3)–(5) show squared values of maximal overlap discrete wavelet transform multi-resolution analysis details of the FYM, KOT and HAG stations, respectively. The window width of eight samples was used to calculate the wavelet coefficients, therefore the length of the detail segments are 1/8 of the length of the original time series. We can recognize from the amplitudes of these MODWT squared details that the first detail in each of these stations had the least amplitude in comparison with the second and third details.

In addition to the segments checked in Figs. (3)–(5) and due to the changing behavior of seismic noise, a study was done on the noise recorded at the three stations to ensure the homogeneous distribution of power over frequency and during different seasons of the year.

Using these noise segments, we calculated:

$$\tilde{\mathbf{V}}_{av}(j,s) = \left[\frac{1}{N} \sum_{n=1}^N \tilde{v}_{j,s,n}^2 \right]$$

where $\tilde{\mathbf{V}}_{av}(j,s)$ is the squared detail average matrix. Elements of this matrix are averages of each MODWT squared detail of each

noise segment, j is the MODWT stage number, which equal to three in this case. $s=1,2,3,4,\dots,S$ and S is the number of noise segments, 64 segments per station.

The standard deviation, $STD(j)$, of each squared detail averages was calculated to judge the distribution of these averages.

$$STD(j) = \sqrt{\sum_{s=1}^S \left(\tilde{\mathbf{V}}_{av}(j,s) - \frac{\overline{\tilde{\mathbf{V}}_{av}(j,s)}}{S-1} \right)^2}$$

The procedure for calculating the squared detail average matrix and standard deviation values of each detail was repeated for each of the three stations under inspection. Table 1 shows these values. From these values, we can conclude the following:

1. The standard deviation averages of the squared first detail are lower than those of the second and third details. This is an indicator that building a *P*-wave arrival detection algorithm based on first detail will not face varieties in noise powers that in turn add robustness to the proposed algorithm.
2. The power level of the FYM station is higher than that of the HAG station which is due to the type of ground on which each station is installed. The power of the noise segments recorded at the FYM station are amplified because this station is installed on soil, while the power of noise segments recorded at the HAG station is less because HAG is on bedrock. Regarding KOT station the STD values are a little bit high because

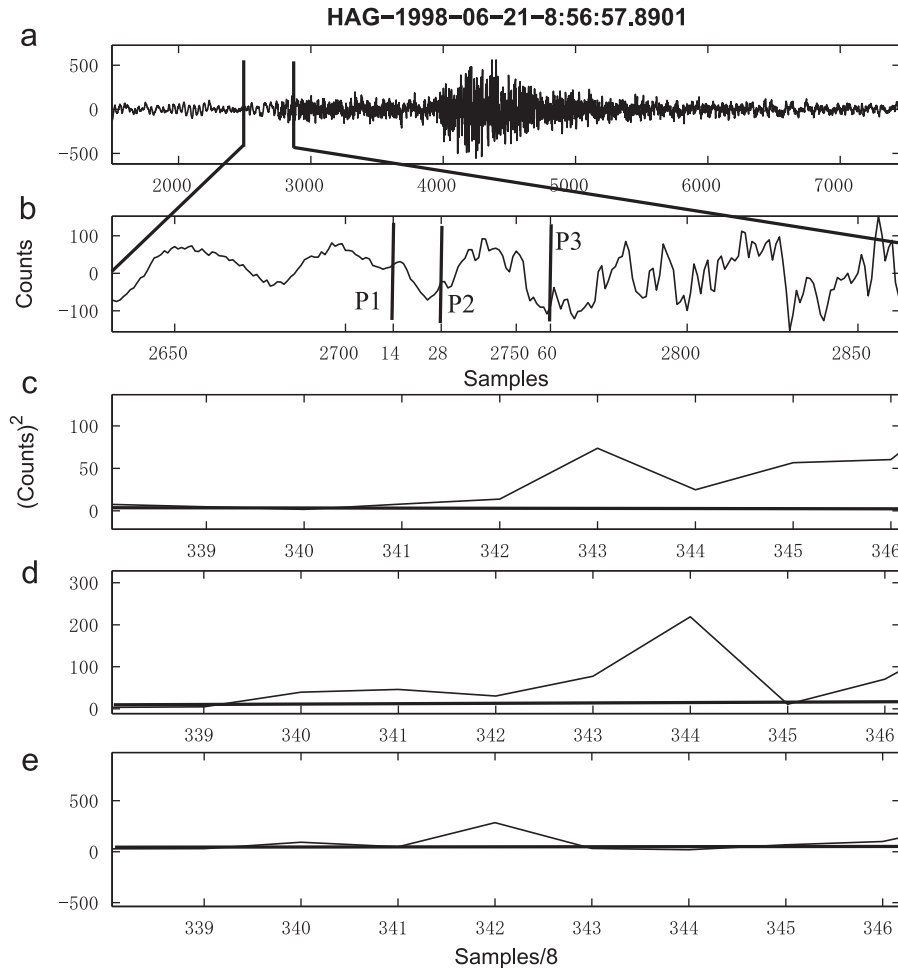


Fig. 7. The first detail amplitudes are apparently greater than the threshold, this feature can not be found in the other details. (a) Time series of an earthquake recorded at Hag station. (b) *P*-wave arrival of the original time series for that earthquake where the three pickings (P1, P2 and P3) which are based on the reference, first and second details are shown, respectively. (c) *P*-wave arrival of the first stage detail. (d) *P*-wave arrival of the second stage detail. (e) *P*-wave arrival of the third stage detail. The bold horizontal line in (b, c, d and e) is the maximum noise level recorded at the start of each segment.

there is low and high frequency noise exists in the record which is clear in Fig. (4).

4. MODWT for manual weak *P*-wave arrival detection

The results of the investigations made on seismic noise in previous section show that the MODWT first square detail has less noise than the other details. In this section, we use this criterion to develop a tool that supports the manual detection of weak onset arrivals. Hafez et al. (2010) explored this topic using discrete wavelet transform (DWT). The same weak event used by Hafez et al. (2010) is checked by the proposed MODWT details. Fig. 6(a), shows the whole weak event to the right and the expected interval that contains the *P*-wave arrival to the left. The arrival timing of this *P*-wave could not be determined manually during the routine work of the seismologists because of the weak arrival. Thus, we can not find any information about this earthquake in National Research Institute of Astronomy and Geophysics (NRIAG) (2002). With the help of the MODWT squared first detail shown in Fig. 6(b), accurate *P*-wave arrival timing can be recognized at sample number 798, which corresponds to sample number 6,384 of the original time series. The second and third squared details have more noise, which hinders accurate detection of *P*-wave arrival, as shown in Fig. 6(c) and (d).

In Hafez et al. (2010), DWT was used to introduce the *P*-wave picking algorithm. That algorithm found that Haar wavelet filter gives the best results, in comparison with other wavelet filters. In order to build an algorithm based on MODWT we use three wavelet filters for comparison; the Haar, Daubechies wavelets of order 4 and 8, (db4) and (db8), Daubechies (1992). A short time-window is required for more accurate *P*-wave arrival determination. The picking accuracy of the algorithm is affected by this window length. Where longer window will decrease the accuracy and shorter window will decrease the resolution of the of the frequency representation. The window width of eight samples was used again to calculate the wavelet coefficients. Using these parameters, we obtained smooth and detail features at three scales (stages). Checking the three details obtained, we found that the first stage detail has the least minimum noise level and the clearest *P*-wave arrival.

5. MODWT for automatic *P*-wave arrival detection

The squared value of first stage detail of MRA of the MODWT is the parameter or criterion that is monitored to check whether or not there is a *P*-wave in that segment and to calculate the timing of onset. This parameter has four important characteristics: (i) The noise level is much lower than the event level, as proven in the seismic noise study conducted in Section 3. (ii) All of its

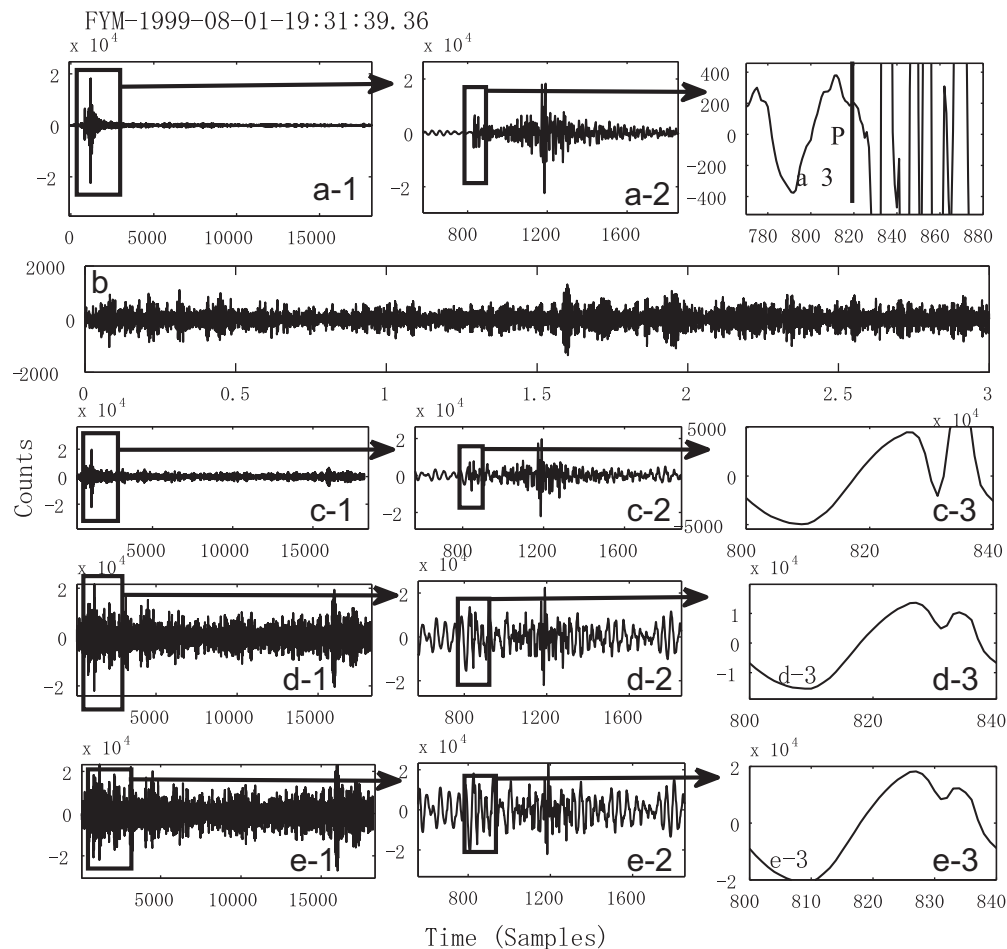


Fig. 8. Pg wave contaminated with different levels of seismic noise. a-1, a-2 and a-3 are an original event recorded at FYM station with (ML=1) and two enlarged views of Pg wave, respectively. The *P*-wave arrival time is marked as *P* b is a seismic noise segment recorded at FYM station also. c-1, d-1 and e-1 are the result of adding the scaled noise in b to a-1 to give SNR=10, 3 and 0, respectively. c-2, d-2 and e-2 are the enlarged views of *P*-wave in c-1, d-1 and e-1, respectively. c-3, d-3 and e-3 are the enlarged views centered around the *P*-wave in c-2, d-3 and e-4, respectively.

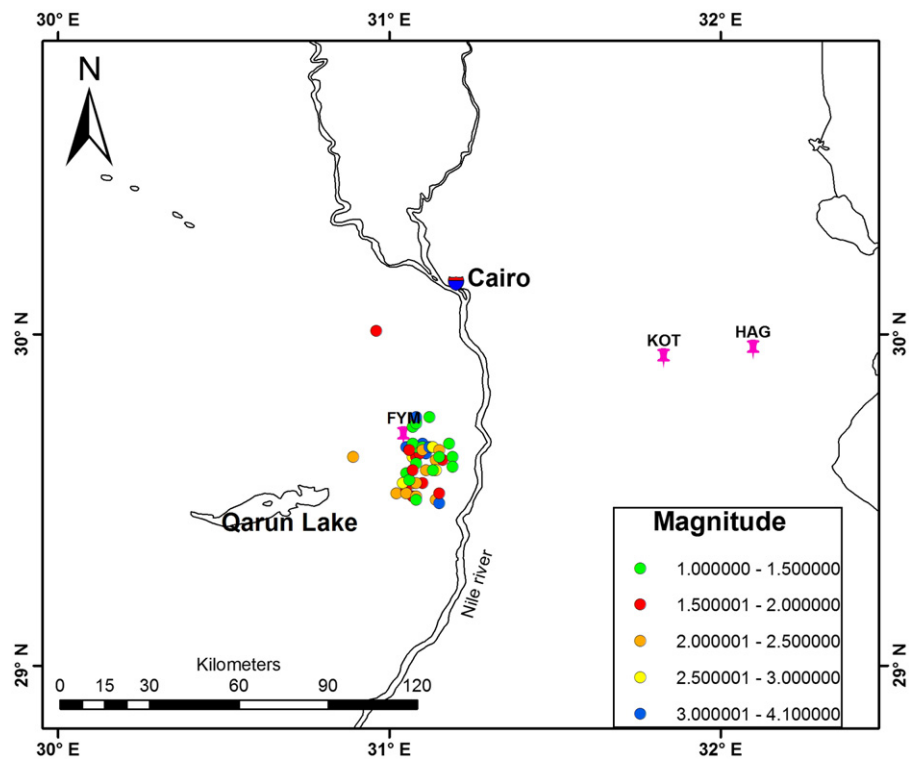


Fig. 9. Map of Cairo region showing the location and local magnitude of the events recorded by FYM, HAG and KOT stations.

Table 2
Percentage of the correct picks for the three details. in each of these percentages, values of correct picks are calculated in successive ranges of seconds.

Detail		Correct picks %	0.0–0.12 s.%	0.13–0.22 s.%	0.23–0.7 s.%	Standard deviation	Average error
Haar	First	87	75.6	15	9.4	0.17	0.02
	Second	85.6	77.6	11.2	11.2	0.16	0.02
	Third	74.6	72.4	9.3	18.3	0.16	0.01
Dua4	First	87.6	75.7	13.2	11.1	0.16	0.01
	Second	86.3	78.5	11.1	10.4	0.15	0.02
	Third	75.3	72.7	9.1	18.2	0.17	0.06
Dau8	First	86.3	73	14.2	12.8	0.17	0.02
	Second	86.3	79.3	7.1	13.6	0.14	0.02
	Third	75.3	71.8	17.2	10.9	0.17	0.06

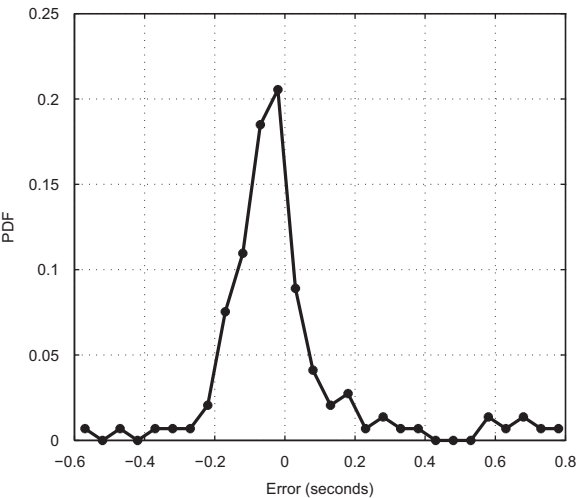


Fig. 10. Probability distribution function of error of the proposed algorithm.

Table 3
Comparison between results obtained when using the MODWT, Spectro-ratio and STA/LTA algorithms with the 146 events.

Algorithm	Correct picks %	0.0–0.12 s.%	0.13–0.22 s.%	0.23–0.7 s.%	Standard deviation
MODWT	87	75.6	15	9.4	0.17
Spectro-ratio	83	76.4	20	3.6	0.11
STA/LTA	76	67.4	19.6	12	0.11

elements are positive values. (iii) Changes in its amplitudes are synchronized with the arrival of the *P*-wave. (iv) An increase in its amplitude at arrival is very clear, making it easy to detect the onset. These features are illustrated in Fig. 7. From Fig. 7 we can find that the accuracy of the picking confirms which detail should be chosen. Here we can recognize that the first detail shows good synchronization than the second detail, where P1 (shown in Fig. 7b) is the reference picking which is at sample 2714. The

picking based on the first detail (P2) is at sample 2728 which shows 14 samples error (0.14 s). The picking based the second detail (P2) is at sample 2760 which means 46 samples error (0.46 s).

The threshold which is chosen to detect the presence of P -wave arrival is denoted by θ . It is defined as the maximum value of the first eight seconds of the chosen squared detail. Calculating that threshold for each segment allows the algorithm to adapt to any changes in seismic noise level in the time just preceding the event.

The selected detail is squared, then monitored: When there are samples that last longer than the maximum recorded duration for the noise specified by [Scherbaum and Johnson \(1992\)](#) and their amplitudes exceed the threshold θ , an event is declared.

The proposed algorithm has some advantages when compared with other algorithms that use MODWT such as [Galiana-Merino et al. \(2008\)](#) as follows;

- (1) Our proposed algorithm uses only one scale detail for automatic picking, which adds simplicity in applying the algorithm to real data networks.
- (2) The proposed algorithm can do picking for a very short window width of up to 40 samples. Kurtosis based algorithm [Galiana-Merino et al. \(2008\)](#) needs a 250–500 samples window width to correctly estimate the first P -wave arrival. This advantage enables our algorithm to be applied in earthquake early warning systems such as [Horiuchi et al. \(2005\)](#), which needs the automatic picking algorithm to make the decision in less than 1 s, meaning that the segment should be less than a 100 sample window width in the case of a 100 samples per second sampling rate.

- (3) Our proposed algorithm is very sensitive to low SNR segments, as will be discussed in next section.

6. Simulation results and comparison

6.1. SNR check

In this section the proposed algorithm is tested on a real P_g wave contaminated with different seismic noise levels to provide different signal to noise ratios, SNRs. This check demonstrates the capability of the algorithm to detect very low SNRs. This P -wave is of an earthquake with a local magnitude of 1 ML recorded at FYM station. The noise segment used, also from FYM station, is scaled by different factors to generate SNRs equals (10, 3 and 0 dB) as shown in [Fig. 8](#). The arrival timing of the P -wave calculated by analyst at NRIAG is at sample 816. Using our proposed algorithm for the original signal, it gave the exact same picking, at sample 816. Adding scaled noise segments and checking arrival timings calculated by the proposed algorithm also gave the picking at sample 816. Thus, we can conclude that the proposed algorithm is very sensitive to very low SNR segments where the event can hardly be recognized from seismic noise, as shown in [Fig. 8\(d-1,2,3 and e-1,2,3\)](#).

7. Application; real data set

We tested our proposed picker and compared it with manual picks for 146 local earthquakes from the Cairo region detected by

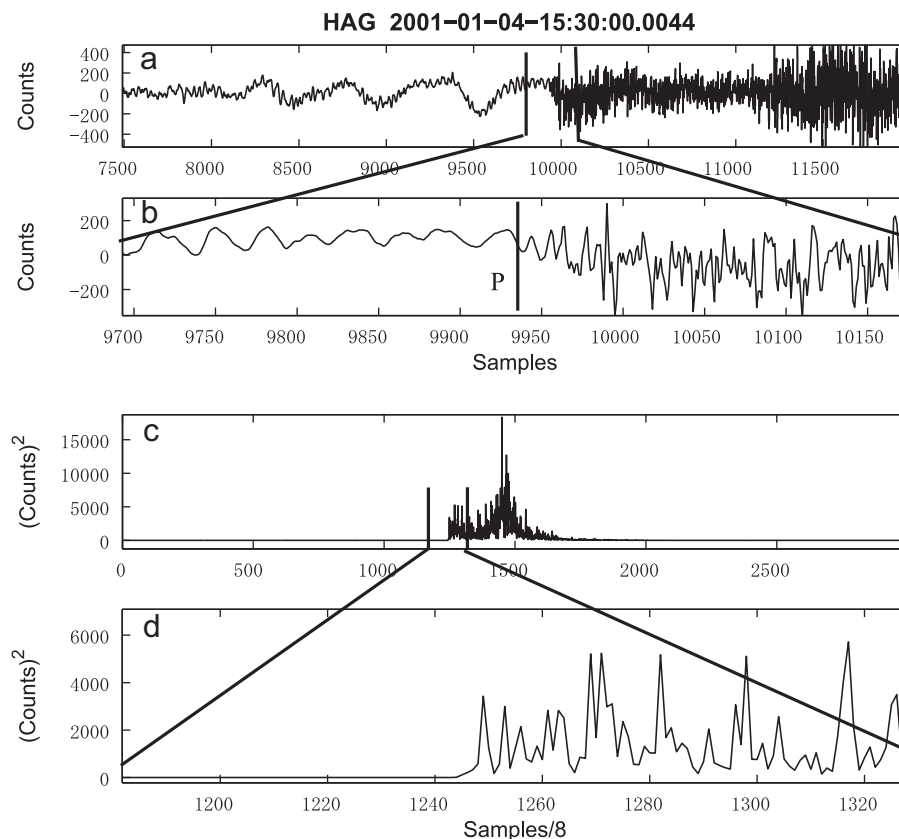


Fig. 11. Although making the decision of the P -wave is difficult in time domain, the first detail gives a clear transition between the noise and the P -wave. (a) Complete event. (b) First arrival is enlarged where the P -wave arrival time is marked as P . (c) Squared first stage MODWT detail of the event. (d) First arrival of the squared first stage is enlarged.

the FYM, HAG and KOT (ENSN stations). The manual picks for these earthquakes were taken from the [National Research Institute of Astronomy and Geophysics \(NRIAG\) \(2002\)](#). These earthquakes were of different magnitudes and occurred at different times to insure that they sustained different seismic noise conditions. The local magnitudes of these earthquakes ranged from 1 to 4.1, as shown in [Fig. 9](#). The 146 segments are short period and very broadband records. The data set used is the same set used by [Hafez et al. \(2009, 2010\)](#).

MODWT details were calculated for each of the eight samples using Haar and Daubechies wavelets of order 4 (db4) and 8 (db8). To confirm the results of seismic noise analysis made in section III, the three wavelet details obtained were used to determine *P*-wave arrival timing. The first detail gave the best results for the wavelet types used, as illustrated in [Table 2](#). Comparing the proposed algorithm with manual picks, we found that both Haar and Daubechies wavelets gave good results. The Haar wavelet is preferred because it has a number of advantages, including conceptual simplicity, speed and memory efficiency. [Fig. 10](#), shows the probability distribution function of the error using Haar wavelet.

Correct picks means the picks are within 0.7 s of the analyst pick. Each range of seconds contains a percent of this correct pick. For example; when using the first detail of Haar wavelet:

Correct picks were 87% meaning that the algorithm gave an error less than 0.7 s for 127 of the 146 events.

Within these 127 events; 75.6% (96 events) were between 0 and 0.12 s.

15% (19 events) were between 0.13 and 0.22 s.

9.4% (12 events) were between 0.23 and 0.7 s.

The algorithm is a statistical model, which generates random error. Standard deviation and statistical averaging of the error are measures of the robustness of this algorithm. The calculated statistical average error was 0.02 s when using the first detail of Haar wavelet filter, as there are positive and negative error values.

[Table 3](#), contains a comparison between the results obtained using the MODWT based algorithm introduced in this section and the results using the spectro-ratio algorithm of [Hafez et al. \(2009\)](#) and the STA/LTA algorithm of [Amadej \(2002\)](#). These results show that the accuracy of the proposed algorithm is much better compared to other algorithms which are already used in seismic networks, where we can derive from the first two columns in [Table 3](#) that the proposed algorithms made a high percentage of the over all correct picks and the percentage of the accurate range of picks as well.

We used the same event shown in [Hafez et al. \(2009\)](#) to check the arrival using maximal overlap discrete wavelet transform as shown in [Fig. 11](#). The change in the power of the selected MODWT detail is clear. The error in this event using our proposed algorithm is 0.02 s. When using the Spectro-ratio and STA/LTA algorithms, the error is 0.05 and 0.18 s, respectively.

8. Conclusion

A multi resolution analysis first detail is monitored and searched for successive samples that have power greater than the adaptive threshold. This leads us to the arrival of onset timing. We used the chosen detail as a tool to determine the *P*-wave arrival manually for weak events. Numerical simulations elaborated that the proposed algorithm is very sensitive to small changes of power in the first arrival of a *P*-wave. Comparison of the time of arrival computed by our proposed algorithms and that

published by NRIAG showed that the error of our algorithm is insignificant when used to calculate earthquake parameters.

Acknowledgments

We would like to express our deepest gratitude and sincere appreciation to the staff in the Egyptian National Seismic Network (ENSN) for their help in preparing the data. We are also grateful to the Egyptian government for the complete support of Mr. Ali G. Hafez during his graduate studies.

Appendix A1

The response of a seismometer to ground motion not only depends on the amplitude of the ground motion (how large it is) but also on its time scale (how sudden it is), or in different words on its frequency.

The seismometers are classified into one of four categories according to their bandwidth:

1. Short Period (SP) seismometers.

The natural frequency of this type is typically 1–2 Hz, and the bandwidth of the recorded signal is between 1 and 50 Hz.

2. Long Period (LP) seismometers

The natural frequency of this type is typically 20–30 mHz, and the bandwidth of the recorded signal is between 0.02 and 10 Hz.

3. Broad Band (BB) seismometers

The natural frequency of this type is typically 20–30 mHz, and the bandwidth of the recorded signal is between 0.02 and 50 Hz.

4. Very Broad Band (VBB) seismometers.

The natural frequency of this type is typically 8 mHz, and the bandwidth of the recorded signal is between 0.008 and 50 Hz.

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